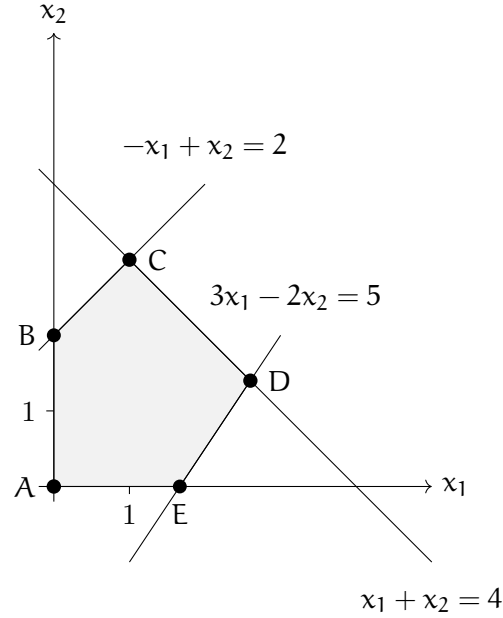


1. The feasible set is shown on the Figure below.



The set of vertices is

	(x_1, x_2)	$-3x_1 - 4x_2$
A	(0,0)	0
B	(0,2)	-8
C	(1,3)	-15
D	(13/5, 7/5)	-13.4
E	(5/3,0)	-5.

Calculating the objective function at each vertex, we obtain that C(1,3) is optimal with optimal value -15. Note that, at the optimal solution the constraints

$$\begin{aligned} -x_1 + x_2 &\leq 2, \\ x_1 + x_2 &\leq 4, \end{aligned}$$

are active, while the constraint

$$-3x_1 + 2x_2 \geq -5$$

is not.

2. Using the summary tables, we obtain the dual problem:

$$\max 2\gamma_1 - 5\gamma_2 + 4\gamma_3$$

subject to

$$\begin{aligned} -\gamma_1 - 3\gamma_2 + \gamma_3 &\leq -3, \\ \gamma_1 + 2\gamma_2 + \gamma_3 &\leq -4, \\ \gamma_1 &\leq 0, \\ \gamma_2 &\geq 0, \\ \gamma_3 &\leq 0. \end{aligned}$$

3. By complementarity slackness, as the second constraint is not active, the corresponding dual variable must be zero: $\gamma_2^* = 0$. Also, as the two primal variables are non zero, the corresponding dual constraints must be active, and we have the following equations:

$$\begin{aligned} -\gamma_1^* - 3\gamma_2^* + \gamma_3^* &= -3, \\ \gamma_1^* + 2\gamma_2^* + \gamma_3^* &= -4. \end{aligned}$$

Putting the two together, we have

$$\begin{aligned} -\gamma_1^* + \gamma_3^* &= -3, \\ \gamma_1^* + \gamma_3^* &= -4, \end{aligned}$$

so that the optimal solution of the dual is

$$\begin{aligned} \gamma_1^* &= -1/2, \\ \gamma_2^* &= 0, \\ \gamma_3^* &= -7/2. \end{aligned}$$

The optimal value is

$$2\gamma_1^* - 5\gamma_2^* + 4\gamma_3^* = -1 - 14 = -15,$$

as expected from the strong duality theorem.